

Global Polynomial Synchronization of Uncertain Complex-Valued Reaction–Diffusion T–S Fuzzy Memristive Neural Networks With Proportional Delays Under Adaptive Event-Triggered Control

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Abstract—This article addresses global polynomial synchronization (GPS) for uncertain complex-valued reaction–diffusion Takagi–Sugeno (T–S) fuzzy memristive neural networks (MNNs) with proportional delays. First, the interval matrix theory is extended, for the first time, to achieve the characterization of complex-domain memristors. Second, novel Lyapunov functionals are constructed by exploiting conjugate transpose properties, circumventing the decomposition of complex states into real and imaginary parts. Subsequently, two event-triggered controllers are designed to derive synchronization criteria, further considering the case where GPS degenerates into global asymptotic synchronization (GAS). Finally, the validity of the theoretical results is verified through two numerical examples.

Index Terms—Event-triggered controllers, global polynomial synchronization (GPS), interval matrix theory, memristive neural networks (MNNs), proportional delay, reaction–diffusion, Takagi–Sugeno (T–S) fuzzy.

$e(\varrho, t) \in \mathbb{C}^n$ $(e_1(\varrho, t), e_2(\varrho, t), \dots, e_n(\varrho, t))^T$.
 $\|\cdot\|$ and $\|A\|$ $(\|\bar{p}\|)^{1/2}$ and $(\mathcal{T}_{\max}(A^H A))^{1/2}$ (matrix A).
 $\|e(\varrho, t)\|$ $(\int_{\Omega} \sum_{k=1}^n e_k(\varrho, t) \overline{e_k(\varrho, t)} d\varrho)^{1/2}$.
 $\varphi_k(\varrho, t) \in \mathbb{C}$ State function.
 $\beta_{ks} > 0$ Transport-diffusion coefficient.
 $d_k^{(p)} > 0$ Fuzzy self-feedback coefficient.
 $q_i \in (0, 1)$ Proportional delay factor.
 $\hat{q} = \min_{i \in \mathcal{N}} \{q_i\}$
 $q_i t = t - (1 - q_i)t$, with $\lim_{t \rightarrow +\infty} (1 - q_i)t = +\infty$ Proportional delay function and its unboundedness.
 $f_i(\cdot), g_i(\cdot)$ Activation functions.
 $b_{ki}(\cdot), c_{ki}(\cdot)$ Memristive weights.
 $\Delta b_{ki}(\varrho, t), \Delta c_{ki}(\varrho, t)$ Uncertain parameters.
 $\varpi_k(\varrho, t)$ Initial functions.

NOMENCLATURE

$\mathcal{N}$ and $i = (-1)^{1/2}$	$\{1, 2, \dots, n\}$ and imaginary unit.
$\mathcal{R}(\mathbb{C})$	Real (complex) domain.
$\mathcal{R}^{n_1 \times n_2} (\mathbb{C}^{n_1 \times n_2})$	$n_1 \times n_2$ real (complex) matrix space.
$\mathcal{R}^m (\mathbb{C}^m)$	Real (complex) vector of dimension m .
$\mathbf{p}$ and $\bar{\mathbf{p}}$	$u_1 + u_2 i$ and $u_1 - u_2 i$, with $u_1, u_2 \in \mathcal{R}$.
$\text{Re}(\mathbf{p}) = 1/2(\mathbf{p} + \bar{\mathbf{p}})$	Real part of $\mathbf{p}$.
$\text{Im}(\mathbf{p}) = 1/2i(\mathbf{p} - \bar{\mathbf{p}})$	Imaginary part of $\mathbf{p}$.
$\Xi^{\uparrow}\{v, \varsigma\}, \Xi^{\downarrow}\{v, \varsigma\}$	$\max\{\text{Re}(v), \text{Re}(\varsigma)\}, \min\{\text{Re}(v), \text{Re}(\varsigma)\}$.
$\Phi^{\uparrow}\{v, \varsigma\}, \Phi^{\downarrow}\{v, \varsigma\}$	$\max\{\text{Im}(v), \text{Im}(\varsigma)\}, \min\{\text{Im}(v), \text{Im}(\varsigma)\}$.
Ω	Bounded domain with smooth boundary Γ and positive measure ($\text{mes}(\Omega) > 0$).
$\mathcal{C} \triangleq \mathcal{C}(\Omega)$	Banach space of continuous functions.
$\mathcal{T}_{\max}(\cdot)$	Largest eigenvalue of matrix.
$\text{diag}(\cdot)$	Diagonal matrix.
$A^T (A^H)$	Transpose (conjugate transpose) of A .

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I. INTRODUCTION

MEMRISTOR is an electronic device with nonvolatile memory characteristics. Chua [1] proposed the theoretical framework of memristors based on considerations of symmetry and completeness in circuit theory, establishing a mathematical relationship between electric charge and magnetic flux. As a frontier interdisciplinary field at the intersection of memristors and neural networks (NNs), memristive NNs (MNNs) enable more efficient and biologically plausible intelligent computing paradigms, closely emulating biological brains.

Due to delays in signal propagation and information processing, delay is inevitable, and its negative impacts can only be predicted, compensated for, or minimized through various methods. Consequently, the study of MNNs with delays holds greater practical significance. Thus, examining dynamic behaviors such as stability [2] and passivity [3], along with behavioral phenomena like synchronization [4], [5], in delayed MNNs has heightened practical relevance. Synchronization, which refers to the process by which two or more NNs with distinct dynamical behaviors gradually achieve consistent states through coupling or external control, holds critical value in fields such as engineering control and secure communication. Common controllers for synchronization include adaptive controller, intermittent controller, and event-triggered controller [4], [5], [6], [7], [8]. The event-triggered approach

updates control only when system states meet predefined triggering conditions. This state-driven strategy provides efficient NNs synchronization in resource-constrained environments. A data-driven event-triggered control scheme was designed, establishing a data-based discrete-time NNs system representation from perturbation-corrupted state-input data and deriving a data-based stability criterion in [6]. Zhang et al. [7] designed a memristor-dependent adaptive event-triggered controller for Markovian memristive reaction–diffusion NNs (RDNNs) with partially unknown transition probabilities, thereby reducing network traffic and establishing new stability criteria. The secure control problem for stochastic networked nonlinear systems under DoS attacks and an unknown dead zone is addressed by an adaptive NN-based event-triggered strategy in [8]. However, research on synchronization of MNNs under event-triggered control remains highly limited.

Unlike traditional NNs, RDNNs are built upon continuous partial differential equations, where the reaction term simulates local neuronal dynamics and the diffusion term models spatial information propagation, fundamentally characterizing the dynamic evolution of spatiotemporal continuous fields [9], [10], [11], [12], [13]. Wang and Zhang [9] explored the global exponential synchronization (GES) of competitive reaction–diffusion MNNs with infinite delays by feedback control. In [12] and [13], GES for coupled delayed reaction–diffusion MNNs and finite-time synchronization (FTS) for inertial reaction–diffusion MNNs were investigated through pinning control design. While conventional NNs focus on temporal dynamics, reaction–diffusion models must account for both time evolution and spatial propagation. The introduction of spatial dependence in RDNNs renders their dynamic analysis notably more complex.

Beyond RDNNs, fuzzy NNs (FNNs) integrate fuzzy logic with NNs to overcome limitations of traditional fuzzy systems by combining knowledge representation and self-learning parameter optimization [14], [15]. The Takagi–Sugeno (T–S) variant [16], as the dominant FNNs implementation, is characterized by its fundamental innovation: defining rule conclusions as linear functions of inputs. Owing to computational efficiency and mathematical tractability, it is widely applied in complex system modeling and nonlinear control [17], [18], [19], [20], [21], [22]. Liu et al. [17] studied GES for a class of reaction–diffusion T–S fuzzy MNNs with bounded time-varying delays via the design of adaptive intermittent controllers. Zhang et al. [20] explored synchronization of reaction–diffusion T–S FNNs with bounded time-variant delays by designing two fuzzy sampled-data controllers. The FTS of T–S fuzzy MNNs with constant delays has been investigated, with feedback controllers and pinning controllers separately designed in [21] and [22]. However, reaction–diffusion T–S FNNs are an extension of the RDNN framework. Research on its dynamics has been limited in existing literature, offering substantial opportunities for further investigation.

Most existing studies concentrate on the dynamic analysis of deterministic NNs with bounded time-varying delays. Nevertheless, parametric inaccuracies in modeling, unmodeled dynamics, and external stochastic disturbances collectively introduce multiple uncertainty sources [23], [24]. Furthermore,

proportional delay, recognized as a significant category of unbounded delay, was initially formalized in NNs by Zhou [25] through establishing mathematical models. In contrast to constant and bounded time-varying delays, the intrinsic unbounded characteristic of proportional delay poses fundamental challenges to dynamic behavior analysis [26], [27], [28], [29], [30], [31]. Zhou and Zhao [27] investigated the synchronization for recurrent NNs with proportional delays, first proposing the concept of global polynomial synchronization (GPS) and establishing its relationship with GES. GPS of MNNs under sliding mode control, and projective synchronization of competitive MNNs via feedback control, both featuring proportional delays and uncertain parameters, in [30] and [31]. Although significant attention has been paid to the dynamical behaviors of proportional delay NNs, synchronization via event-triggered controllers has yet to be explored for such networks with integrated T–S fuzzy mechanisms and reaction–diffusion terms, primarily due to the unbounded nature of proportional delays, which introduces fundamental analytical challenges.

Contemporary NNs research continues to prioritize real-valued NNs (RVNNs), whereas complex-valued NNs (CVNNs) constitute their natural mathematical extension. Native operation in the complex domain allows CVNNs to inherently decouple signal amplitude and phase dynamics while enabling linear rotational transformations, consequently establishing a foundational framework for physics-aware intelligent computing [32], [33], [34], [35]. Probe synchronization of uncertain discrete-time fractional-order complex-valued T–S FNNs with bounded time-variant delays was analyzed in [33]. Although CVNNs exhibit unique advantages in processing complex domain data, significant limitations persist at both theoretical and practical levels. For instance, a systematic framework for Lyapunov stability theory in the complex domain remains underdeveloped, and complex differential inclusion theory is yet to be fully established. Under complex constraints such as proportional delays and fuzzy rules, novel analytical tools for the complex domain must be developed to achieve substantive advances.

To more effectively extend the proportional delay framework to CVNNs, T–S FNNs, and RDNNs, this work investigates GPS in proportional delay uncertain complex-valued reaction–diffusion T–S fuzzy MNNs using event-triggered control. Comparisons between the methods used in this article and those reported in the literature are presented in Tables I and II. However, the following critical issues arise: 1) how to design effective adaptive event-triggered controllers while avoiding Zeno behavior? 2) how to handle complex-valued memristive weights more effectively? and 3) how to address the unboundedness of proportional delays? Building on insights from [2], [5], and [17], the contributions of this work are as follows.

- 1) Unlike prior methods applying interval matrix techniques to memristive RVNNs [2], [37], this work first extends these techniques to CVNNs. This avoids the complex separation of real and imaginary parts, yielding a simpler structure.
- 2) Compared with conventional event-triggered controllers [4], [5], our adaptive event-triggered controller merges

the benefits of adaptive and event-triggered control, achieving higher energy efficiency and enhanced stability.

- 3) Compared to analyses of reaction-diffusion T-S FNNs with constant or bounded time-varying delays [18], [20], this article establishes a framework based on proportional delays, thus significantly extending the research scope.

II. PRELIMINARIES

A. Model Dynamics

The following describes uncertain complex-valued reaction-diffusion T-S fuzzy MNNs with proportional delays under initial and boundary conditions.

Fuzzy Rule p: For $\kappa \in \mathcal{N}$, if $\mu_1(t)$ is $\chi_1^{(p)}$, $\mu_2(t)$ is $\chi_2^{(p)}$, and $\mu_Q(t)$ is $\chi_Q^{(p)}$, then

$$\left\{ \begin{aligned} \frac{\partial \varphi_\kappa(\varrho, t)}{\partial t} &= \sum_{s=1}^S \frac{\partial}{\partial \varrho_s} \left(\beta_{\kappa s} \frac{\partial \varphi_\kappa(\varrho, t)}{\partial \varrho_s} \right) - d_\kappa^{[p]} \varphi_\kappa(\varrho, t) \\ &+ \sum_{i=1}^n [b_{\kappa i}(\varphi_\kappa(\varrho, t)) + \Delta b_{\kappa i}(\varrho, t)] \\ &\times f_i(\varphi_i(\varrho, t)) + \sum_{i=1}^n [c_{\kappa i}(\varphi_\kappa(\varrho, t)) \\ &+ \Delta c_{\kappa i}(\varrho, t)] g_i(\varphi_i(\varrho, q_i t)), \quad t \geq t_0 \geq 0 \\ \varphi_\kappa(\varrho, t) &= 0, \quad (\varrho, t) \in \Gamma \times [\hat{q}t_0, +\infty) \\ \varphi_\kappa(\varrho, t) &= \varpi_\kappa(\varrho, t), \quad (\varrho, t) \in \Omega \times [\hat{q}t_0, t_0] \end{aligned} \right. \quad (1)$$

where the notations are defined in the Nomenclature section.

After fuzzification, system (1) can be expressed as

$$\begin{aligned} \frac{\partial \varphi_\kappa(\varrho, t)}{\partial t} &= \sum_{p=1}^P \zeta_p(\mu(t)) \left[\sum_{s=1}^S \frac{\partial}{\partial \varrho_s} \left(\beta_{\kappa s} \frac{\partial \varphi_\kappa(\varrho, t)}{\partial \varrho_s} \right) \right. \\ &- d_\kappa^{[p]} \varphi_\kappa(\varrho, t) + \sum_{i=1}^n [b_{\kappa i}(\varphi_\kappa(\varrho, t)) + \Delta b_{\kappa i}(\varrho, t)] \\ &\times f_i(\varphi_i(\varrho, t)) + \sum_{i=1}^n [c_{\kappa i}(\varphi_\kappa(\varrho, t)) + \Delta c_{\kappa i}(\varrho, t)] \\ &\times g_i(\varphi_i(\varrho, q_i t)) \Big] \end{aligned} \quad (2)$$

where

$$\zeta_p(\mu(t)) = \frac{\prod_{j=1}^Q \chi_j^{(p)}(\mu_j(t))}{\sum_{p=1}^P \prod_{j=1}^Q \chi_j^{(p)}(\mu_j(t))} \geq 0 \quad (3)$$

in which $\chi_j^{(p)}(\mu_j(t))$ denotes the membership function of $\mu_j(t)$ in $\chi_j^{(p)}$, with $\sum_{p=1}^P \zeta_p(\mu(t)) = 1$.

The memristor modeling framework is defined as follows:

$$b_{\kappa i}(\cdot) = \begin{cases} b_{\kappa i}^\lambda, & \dagger_{\kappa 1}, \\ b_{\kappa i}^\zeta, & \dagger_{\kappa 2}, \end{cases} \quad c_{\kappa i}(\cdot) = \begin{cases} c_{\kappa i}^\lambda, & \dagger_{\kappa 1} \\ c_{\kappa i}^\zeta, & \dagger_{\kappa 2} \end{cases}$$

in which $\dagger_{\kappa 1}$ and $\dagger_{\kappa 2}$ represent $|\cdot| \leq \mathcal{T}_\kappa$ and $|\cdot| > \mathcal{T}_\kappa$, and $\mathcal{T}_\kappa > 0$ refers to the switching jumps. $b_{\kappa i}^\lambda, b_{\kappa i}^\zeta, c_{\kappa i}^\lambda, c_{\kappa i}^\zeta \in \mathbb{C}$. Next, the interval matrix theory for real-valued memristors [37], [38], [39] will be extended to the complex domain.

For any $\kappa, i \in \mathcal{N}$, let $b_{\kappa i}^\tau = \Xi^\dagger\{b_{\kappa i}^\lambda, b_{\kappa i}^\zeta\}$, $b_{\kappa i}^\tau = \Xi^\dagger\{b_{\kappa i}^\lambda, b_{\kappa i}^\zeta\}$, $b_{\kappa i}^\tau = \Phi^\dagger\{b_{\kappa i}^\lambda, b_{\kappa i}^\zeta\}$, $b_{\kappa i}^\tau = \Phi^\dagger\{b_{\kappa i}^\lambda, b_{\kappa i}^\zeta\}$, $c_{\kappa i}^\tau = \Xi^\dagger\{c_{\kappa i}^\lambda, c_{\kappa i}^\zeta\}$, $c_{\kappa i}^\tau = \Xi^\dagger\{c_{\kappa i}^\lambda, c_{\kappa i}^\zeta\}$, $c_{\kappa i}^\tau = \Phi^\dagger\{c_{\kappa i}^\lambda, c_{\kappa i}^\zeta\}$, $c_{\kappa i}^\tau = \Phi^\dagger\{c_{\kappa i}^\lambda, c_{\kappa i}^\zeta\}$.

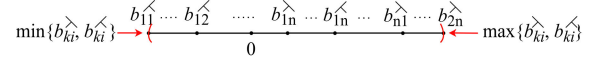


Fig. 1. Value range of real-valued memristive weights.

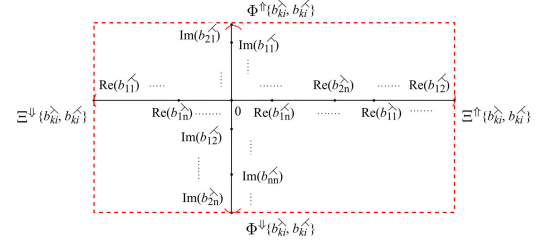


Fig. 2. Value range of complex-valued memristive weights.

$\Xi^\dagger\{c_{\kappa i}^\lambda, c_{\kappa i}^\zeta\}$, $c_{\kappa i}^\tau = \Xi^\dagger\{c_{\kappa i}^\lambda, c_{\kappa i}^\zeta\}$, $c_{\kappa i}^\tau = \Phi^\dagger\{c_{\kappa i}^\lambda, c_{\kappa i}^\zeta\}$, $c_{\kappa i}^\tau = \Phi^\dagger\{c_{\kappa i}^\lambda, c_{\kappa i}^\zeta\}$, and $b_{\kappa i}^\tau = 1/2(b_{\kappa i}^\lambda + b_{\kappa i}^\zeta)$, $b_{\kappa i}^\tau = 1/2(b_{\kappa i}^\lambda - b_{\kappa i}^\zeta)$, $b_{\kappa i}^\tau = 1/2(b_{\kappa i}^\lambda + b_{\kappa i}^\zeta)$, $b_{\kappa i}^\tau = 1/2(b_{\kappa i}^\lambda - b_{\kappa i}^\zeta)$, $c_{\kappa i}^\tau = 1/2(c_{\kappa i}^\lambda + c_{\kappa i}^\zeta)$, $c_{\kappa i}^\tau = 1/2(c_{\kappa i}^\lambda - c_{\kappa i}^\zeta)$, $c_{\kappa i}^\tau = 1/2(c_{\kappa i}^\lambda + c_{\kappa i}^\zeta)$, $c_{\kappa i}^\tau = 1/2(c_{\kappa i}^\lambda - c_{\kappa i}^\zeta)$.

By virtue of Filippov solution and measurable selection theorem [40], system (2) can be reformulated as

$$\begin{aligned} \frac{\partial \varphi(\varrho, t)}{\partial t} &= \sum_{p=1}^P \zeta_p(\mu(t)) \{ \mathbb{E}\{\varphi(\varrho, t)\} - D^{[p]} \varphi(\varrho, t) \\ &+ [B^\Delta + B^\tau + B_1 \Theta_1(t) B_2 + B_3 \Theta_1^\diamond(t) B_4 \\ &+ \Delta B(\varrho, t)] f(\varphi(\varrho, t)) + [C^\Delta + C^\tau + C_1 \Upsilon_1(t) C_2 \\ &+ C_3 \Upsilon_1^\diamond(t) C_4 + \Delta C(\varrho, t)] g(\varphi(\varrho, q_i t)) \} \end{aligned} \quad (4)$$

with $\varphi(\varrho, t) = (\varphi_1(\varrho, t), \dots, \varphi_n(\varrho, t))^T$, $\mathbb{E}\{\varphi(\varrho, t)\} = (\sum_{s=1}^S \partial/(\partial \varrho_s) (\beta_{1s}(\partial \varphi_1(\varrho, t))/(\partial \varrho_s)), \dots, \sum_{s=1}^S \partial/(\partial \varrho_s) (\beta_{ns}(\partial \varphi_n(\varrho, t))/(\partial \varrho_s)))^T$, $f(\varphi(\varrho, t)) = (f_1(\varphi_1(\varrho, t)), f_2(\varphi_2(\varrho, t)), \dots, f_n(\varphi_n(\varrho, t)))^T$, $g(\varphi(\varrho, q_i t)) = (g_1(\varphi_1(\varrho, q_i t)), g_2(\varphi_2(\varrho, q_i t)), \dots, g_n(\varphi_n(\varrho, q_i t)))^T \in \mathbb{C}^n$. $D^{[p]} = \text{diag}(d_\kappa^{[p]})$, $B^\Delta = (b_{\kappa i}^\Delta)_{n \times n}$, $C^\Delta = (c_{\kappa i}^\Delta)_{n \times n} \in \mathbb{R}^{n \times n}$. $B^\tau = (b_{\kappa i}^\tau)_{n \times n}$, $C^\tau = (c_{\kappa i}^\tau)_{n \times n} \in \mathbb{C}^{n \times n}$. $\Theta_1(t) = \text{diag}(\iota_{\kappa i}(t))$, $\Upsilon_1(t) = \text{diag}(\epsilon_{\kappa i}(t)) \in \mathbb{R}^{n^2 \times n^2}$. $\Theta_1^\diamond(t) = \text{diag}(\tilde{\iota}_{\kappa i}(t)i)$, $\Upsilon_1^\diamond(t) = \text{diag}(\tilde{\epsilon}_{\kappa i}(t)i) \in \mathbb{C}^{n^2 \times n^2}$. $B_1, B_3, C_1, C_3 \in \mathbb{R}^{n \times n^2}$. $B_2, B_4, C_2, C_4 \in \mathbb{R}^{n^2 \times n}$.

$$\begin{aligned} B_1 &= (\sqrt{b_{11}^\tau} \vartheta_1, \dots, \sqrt{b_{1n}^\tau} \vartheta_1, \dots, \sqrt{b_{n1}^\tau} \vartheta_n, \dots, \sqrt{b_{nn}^\tau} \vartheta_n) \\ B_2 &= (\sqrt{b_{11}^\tau} \vartheta_1, \dots, \sqrt{b_{1n}^\tau} \vartheta_n, \dots, \sqrt{b_{n1}^\tau} \vartheta_1, \dots, \sqrt{b_{nn}^\tau} \vartheta_n)^T \\ B_3 &= (\sqrt{b_{11}^\tau} \vartheta_1, \dots, \sqrt{b_{1n}^\tau} \vartheta_1, \dots, \sqrt{b_{n1}^\tau} \vartheta_n, \dots, \sqrt{b_{nn}^\tau} \vartheta_n) \\ B_4 &= (\sqrt{b_{11}^\tau} \vartheta_1, \dots, \sqrt{b_{1n}^\tau} \vartheta_n, \dots, \sqrt{b_{n1}^\tau} \vartheta_1, \dots, \sqrt{b_{nn}^\tau} \vartheta_n)^T \end{aligned}$$

where C_1, C_2, C_3, C_4 are defined in a similar manner to B_1, B_2, B_3, B_4 , and ϑ_κ is the standard basis vector with 1 at the κ th position and 0 elsewhere. $\iota_{\kappa i}(t), \tilde{\iota}_{\kappa i}(t), \epsilon_{\kappa i}(t), \tilde{\epsilon}_{\kappa i}(t) \in [-1, 1]$.

Remark 1: Most existing studies that apply interval matrix theory to memristors, typically within real-valued systems, follow a three-step procedure.

- 1) Select the maximum and minimum values of the memristive weights such that all possible values are confined within this range (see Fig. 1). Note that Figs. 1 and 2 are purely schematic. The layout of the memristive weights is arbitrary, depicted solely for illustrative purposes and not indicative of their true numerical values.

- 2) Form a matrix for the element using the midpoint value (the average of the max and min from Step 1) as the nominal value.
- 3) The radius of variation is half the difference between the maximum and minimum values, defining the entire possible interval for the memristance.

A significant challenge arises because the value of a complex-valued memristive weight is confined to a rectangular region (see Fig. 2), unlike the simple line segment for real-valued ones. This makes defining the corresponding variation radius a nontrivial task.

Based on the definitions of B^Δ , B^* , $\mathcal{B}_1$, $\mathcal{B}_2$, $\mathcal{B}_3$, $\mathcal{B}_4$, $\Theta_1(t)$, and $\Theta_1^\diamond(t)$, the resulting complex-valued matrix is given by $[B^\Delta + \mathcal{B}_1\Theta_1(t)\mathcal{B}_2] + [B^* + \mathcal{B}_3\Theta_1^\diamond(t)\mathcal{B}_4] = (b_{ki}^\Delta + b_{ki}^* \iota_{ki}(t))_{n \times n} + (b_{ki}^* + b_{ki}^\Delta \tilde{\iota}_{ki}(t))_{n \times n} i$. In particular, when $\iota_{ki}(t) = -1$ and $\tilde{\iota}_{ki}(t) = -1$, both the real and imaginary parts attain their element-wise minimum values, yielding $(b_{ki}^\Delta)_{n \times n}$ and $(b_{ki}^*)_{n \times n}$, respectively.

When $\iota_{ki}(t) = 1$ and $\tilde{\iota}_{ki}(t) = 1$, both the real and imaginary parts attain their element-wise maximum values, yielding $(b_{ki}^\Delta)_{n \times n}$ and $(b_{ki}^*)_{n \times n}$, respectively.

We regard system (4) as the drive system, and the response system is given as follows:

$$\begin{aligned} \frac{\partial \psi(\varrho, t)}{\partial t} = & \sum_{p=1}^P \zeta_p(\mu(t)) \left\{ \mathbb{K} \{ \psi(\varrho, t) \} - D^{[p]} \psi(\varrho, t) \right. \\ & + [B^\Delta + B^* + \mathcal{B}_1\Theta_2(t)\mathcal{B}_2 + \mathcal{B}_3\Theta_2^\diamond(t)\mathcal{B}_4 \\ & + \Delta B(\varrho, t)] f(\psi(\varrho, t)) + [C^\Delta + C^* \\ & + \mathcal{C}_1\Upsilon_2(t)\mathcal{C}_2 + \mathcal{C}_3\Upsilon_2^\diamond(t)\mathcal{C}_4 + \Delta C(\varrho, t)] \\ & \left. \times g(\psi(\varrho, qt)) + \mathcal{W}(\varrho, t) \right\} \end{aligned} \quad (5)$$

where the initial and boundary conditions are given by $\psi_k(\varrho, t) = \tilde{\omega}_k(\varrho, t)$, $(\varrho, t) \in \Omega \times [\hat{q}t_0, t_0]$ and $\psi_k(\varrho, t) = 0$, $(\varrho, t) \in \Gamma \times [\hat{q}t_0, +\infty)$. $\Theta_2(t) = \text{diag}(\iota_{ki}^*(t))$, $\Upsilon_2(t) = \text{diag}(\epsilon_{ki}^*(t)) \in \mathbb{R}^{n^2 \times n^2}$. $\Theta_2^\diamond(t) = \text{diag}(\tilde{\iota}_{ki}^*(t))$, $\Upsilon_2^\diamond(t) = \text{diag}(\tilde{\epsilon}_{ki}^*(t)) \in \mathbb{C}^{n^2 \times n^2}$. $\iota_{ki}^*(t)$, $\tilde{\iota}_{ki}^*(t)$, $\epsilon_{ki}^*(t)$, $\tilde{\epsilon}_{ki}^*(t) \in [-1, 1]$. $\psi(\varrho, t)$ and $\mathcal{W}(\varrho, t)$ denote the controller. $f(\psi(\varrho, t))$ and $g(\psi(\varrho, qt))$ are defined similar to their counterparts in (4).

Define $e(\varrho, t) = \psi(\varrho, t) - \varphi(\varrho, t) = (\psi_1(\varrho, t) - \varphi_1(\varrho, t), \psi_2(\varrho, t) - \varphi_2(\varrho, t), \dots, \psi_n(\varrho, t) - \varphi_n(\varrho, t))^\top \in \mathbb{C}^n$. With reference to systems (4) and (5), the error system is established as follows:

$$\begin{aligned} \frac{\partial e(\varrho, t)}{\partial t} = & \sum_{p=1}^P \zeta_p(\mu(t)) \left\{ \mathbb{K} \{ e(\varrho, t) \} - D^{[p]} e(\varrho, t) \right. \\ & + [B^\Delta + B^* + \mathcal{B}_1\Theta_2(t)\mathcal{B}_2 + \mathcal{B}_3\Theta_2^\diamond(t)\mathcal{B}_4 \\ & + \Delta B(\varrho, t)] f(e(\varrho, t)) + [\mathcal{B}_1(\Theta_2(t) - \Theta_1(t))\mathcal{B}_2 \\ & + \mathcal{B}_3(\Theta_2^\diamond(t) - \Theta_1^\diamond(t))\mathcal{B}_4] f(\varphi(\varrho, t)) + [C^\Delta + C^* \\ & + \mathcal{C}_1\Upsilon_2(t)\mathcal{C}_2 + \mathcal{C}_3\Upsilon_2^\diamond(t)\mathcal{C}_4 + \Delta C(\varrho, t)] g(e(\varrho, qt)) \\ & \left. + [\mathcal{C}_1(\Upsilon_2(t) - \Upsilon_1(t))\mathcal{C}_2 + \mathcal{C}_3(\Upsilon_2^\diamond(t) - \Upsilon_1^\diamond(t))\mathcal{C}_4] \right. \\ & \left. \times g(\varphi(\varrho, qt)) + \mathcal{W}(\varrho, t) \right\} \end{aligned} \quad (6)$$

under the initial condition $e_k(\varrho, t) = \tilde{\omega}_k(\varrho, t) - \varpi_k(\varrho, t)$, $(\varrho, t) \in \Omega \times [\hat{q}t_0, t_0]$ and boundary condition $e_k(\varrho, t) = 0$, $(\varrho, t) \in \Gamma \times [\hat{q}t_0, +\infty)$. $f(e(\varrho, t)) = f(\psi(\varrho, t)) - f(\varphi(\varrho, t))$, $g(e(\varrho, qt)) = g(\psi(\varrho, qt)) - g(\varphi(\varrho, qt)) \in \mathbb{C}^n$. $\mathbb{K}\{e(\varrho, t)\} = \mathbb{K}\{\psi(\varrho, t)\} - \mathbb{K}\{\varphi(\varrho, t)\}$.

To conserve energy, we implement this event-triggered mechanism

$$t_{r+1} = \inf \{ t > t_r \mid \|e(\varrho, t_r) - e(\varrho, t)\| > \lambda \|e(\varrho, t)\| \} \quad (7)$$

with t_{r+1} being the event-triggered control time sequence, $\lambda \in (0, 1)$.

The underlying assumptions are introduced as follows.

(A₁) Uncertain parameters satisfy that

$$\|\Delta B(\varrho, t)\| \leq \Delta b, \quad \|\Delta C(\varrho, t)\| \leq \Delta c.$$

(A₂) For arbitrary $\mathbf{a}, \mathbf{b} \in \mathbb{C}^n$, the activation functions $f(\cdot)$ and $g(\cdot)$ meet the following criteria:

$$\begin{aligned} \|f(\mathbf{a}) - f(\mathbf{b})\| &\leq F \|\mathbf{a} - \mathbf{b}\|, \quad F \geq 0, \quad \|f(\cdot)\| \leq \mathcal{L}_1 \\ \|g(\mathbf{a}) - g(\mathbf{b})\| &\leq G \|\mathbf{a} - \mathbf{b}\|, \quad G \geq 0, \quad \|g(\cdot)\| \leq \mathcal{L}_2. \end{aligned}$$

Remark 2: The uncertainty assumption (A₁) posits that most disturbances are localized, which prevents system divergence, and the assumption on the activation functions (A₂) requires them to be Lipschitz continuous and bounded, which ensures the existence of solutions to the error system (6).

Definition 1: [27] Systems (4) and (5) are GPS, given the existence of constants $\delta > 0$ and $\aleph \in [1, +\infty)$ for which

$$\|e(\varrho, t)\|^2 \leq \aleph \|\tilde{\omega} - \varpi\|_{\hat{q}}^2 \left(\frac{1+t}{1+t_0} \right)^{-\delta}, \quad t \geq t_0 \geq 0$$

where $\|\tilde{\omega} - \varpi\|_{\hat{q}}^2 = \int_{\Omega} \sum_{k=1}^n \sup_{\mathcal{U}} |\tilde{\omega}_k(\varrho, t) - \varpi_k(\varrho, t)|^2 d\varrho$, $\mathcal{U} = \{(\varrho, t) \mid (\varrho, t) \in \Omega \times [\hat{q}t_0, t_0]\}$.

Definition 2: [27] Systems (4) and (5) are global asymptotic synchronization (GAS) if

$$\lim_{t \rightarrow \infty} \|\varphi(\varrho, t) - \psi(\varrho, t)\| = 0, \quad t \geq t_0 \geq 0.$$

Lemma 1: [36] Let $\Omega = \{\varrho = (\varrho_1, \varrho_2, \dots, \varrho_S)^\top \mid |\varrho_s| < \theta_s, s = 1, 2, \dots, S\} \subset \mathbb{R}^S$ be a polydisc with smooth boundary Γ . For any real-valued function $\phi(\varrho) \in \mathcal{C}$ satisfying $\phi(\varrho)|_{\partial\Omega} = 0$, the inequality holds

$$\int_{\Omega} |\phi|^2 d\varrho \leq \theta_s^2 \int_{\Omega} \left(\frac{\partial \phi}{\partial \varrho_s} \right)^2 d\varrho$$

where $d\varrho$ denotes the standard volume measure.

III. MAIN RESULTS

For establishing GPS between systems (4) and (5), an adaptive event-triggered controller is designed, incorporating polynomial functions and the fuzzy logic (3)

$$\mathcal{W}(\varrho, t) = -\mathcal{W}(\varrho, t) e(\varrho, t_r) - \mathcal{M}(\varrho, t) \frac{e(\varrho, t_r)}{\|e(\varrho, t)\|} \quad (8)$$

with $t \in [t_r, t_{r+1})$, $\mathcal{W}(\varrho, t) = \text{diag}(\mathcal{W}_n(\varrho, t))$, $\mathcal{M}(\varrho, t) = \text{diag}(\mathcal{M}_n(\varrho, t)) \in \mathbb{C}^{n \times n}$. $(\partial \mathcal{W}_k(\varrho, t))/(\partial t) = \sum_{p=1}^P \zeta_p(\mu(t)) (1 + t)^\delta \xi_{1k} e_k(\varrho, t) \overline{e_k(\varrho, t_r)}$, $(\partial \mathcal{M}_k(\varrho, t))/(\partial t) = \sum_{p=1}^P \zeta_p(\mu(t)) (1 + t)^\delta \xi_{2k} e_k(\varrho, t) \|e(\varrho, t)\|^{-1} \overline{e_k(\varrho, t_r)}$. $\xi_{1k}, \xi_{2k} > 0$ are adaptive control parameters and $\delta > 0$.

Theorem 1: Under (A₁) and (A₂), GPS between systems (4) and (5) is attainable via adaptive event-triggered controller (8).

Proof: The Lyapunov functional takes the composite form

$$\mathcal{V}(\varrho, t) = \mathcal{V}_1(\varrho, t) + \mathcal{V}_2(\varrho, t) + \mathcal{V}_3(\varrho, t) \quad (9)$$

with

$$\begin{aligned} \mathcal{V}_1(\varrho, t) &= \int_{\Omega} (1+t)^{\delta} e^H(\varrho, t) e(\varrho, t) d\varrho \\ \mathcal{V}_2(\varrho, t) &= 5G^2 \int_{\Omega} \sum_{i=1}^n \int_{q,t}^t (1+q_i^{-1}s)^{\delta} q_i^{-1} |e_i(\varrho, s)|^2 ds d\varrho \\ \mathcal{V}_3(\varrho, t) &= \int_{\Omega} \sum_{\kappa=1}^n [\xi_{1\kappa}^{-1} (\overline{\mathcal{W}_{\kappa}(\varrho, t) - \mathcal{W}_{\kappa}}) (\mathcal{W}_{\kappa}(\varrho, t) - \mathcal{W}_{\kappa}) \\ &\quad + \xi_{2\kappa}^{-1} (\overline{\mathcal{M}_{\kappa}(\varrho, t) - \mathcal{M}_{\kappa}}) (\mathcal{M}_{\kappa}(\varrho, t) - \mathcal{M}_{\kappa})] d\varrho \end{aligned}$$

where $\mathcal{W}_{\kappa}, \mathcal{M}_{\kappa} \in \mathbb{C}$.

For any $0 \leq t_0 \leq t$, computing the derivative of $\mathcal{V}_1(\varrho, t)$ along (6) yields

$$\begin{aligned} \frac{\partial \mathcal{V}_1(\varrho, t)}{\partial t} &= \int_{\Omega} \left[\delta (1+t)^{\delta-1} e^H(\varrho, t) e(\varrho, t) + \sum_{p=1}^P \zeta_p(\mu(t)) \right. \\ &\quad \left. \times (1+t)^{\delta} \left[e^H(\varrho, t) \frac{\partial e(\varrho, t)}{\partial t} + \left(\frac{\partial e(\varrho, t)}{\partial t} \right)^H e(\varrho, t) \right] \right] d\varrho. \end{aligned} \quad (10)$$

By substituting (6) into (10), we obtain

$$\begin{aligned} &\int_{\Omega} \left[e^H(\varrho, t) \frac{\partial e(\varrho, t)}{\partial t} + \left(\frac{\partial e(\varrho, t)}{\partial t} \right)^H e(\varrho, t) \right] d\varrho \\ &= \int_{\Omega} \left[e^H(\varrho, t) \mathbb{K}\{e(\varrho, t)\} + \mathbb{K}^H\{e(\varrho, t)\} e(\varrho, t) \right. \\ &\quad - 2e^H(\varrho, t) D^{[p]} e(\varrho, t) + e^H(\varrho, t) ([B^{\Delta} + B^* + \mathcal{B}_1 \Theta_2(t) \mathcal{B}_2 \\ &\quad + \mathcal{B}_3 \Theta_2^{\diamond}(t) \mathcal{B}_4 + \Delta B(\varrho, t)] f(e(\varrho, t)) + [\mathcal{B}_1(\Theta_2(t) \\ &\quad - \Theta_1(t)) \mathcal{B}_2 + \mathcal{B}_3(\Theta_2^{\diamond}(t) - \Theta_1^{\diamond}(t)) \mathcal{B}_4] f(\varphi(\varrho, t)) + [C^{\Delta} \\ &\quad + C^* + \mathcal{C}_1 \Upsilon_2(t) \mathcal{C}_2 + \mathcal{C}_3 \Upsilon_2^{\diamond}(t) \mathcal{C}_4 + \Delta C(\varrho, t)] g(e(\varrho, qt)) \\ &\quad + [\mathcal{C}_1(\Upsilon_2(t) - \Upsilon_1(t)) \mathcal{C}_2 + \mathcal{C}_3(\Upsilon_2^{\diamond}(t) - \Upsilon_1^{\diamond}(t)) \mathcal{C}_4] \\ &\quad \times g(\varphi(\varrho, qt)) + \mathcal{U}(\varrho, t) + ([B^{\Delta} + B^* + \mathcal{B}_1 \Theta_2(t) \mathcal{B}_2 \\ &\quad + \mathcal{B}_3 \Theta_2^{\diamond}(t) \mathcal{B}_4 + \Delta B(\varrho, t)] f(e(\varrho, t)) + [\mathcal{B}_1(\Theta_2(t) \\ &\quad - \Theta_1(t)) \mathcal{B}_2 + \mathcal{B}_3(\Theta_2^{\diamond}(t) - \Theta_1^{\diamond}(t)) \mathcal{B}_4] f(\varphi(\varrho, t)) + [C^{\Delta} \\ &\quad + C^* + \mathcal{C}_1 \Upsilon_2(t) \mathcal{C}_2 + \mathcal{C}_3 \Upsilon_2^{\diamond}(t) \mathcal{C}_4 + \Delta C(\varrho, t)] g(e(\varrho, qt)) \\ &\quad \left. + [\mathcal{C}_1(\Upsilon_2(t) - \Upsilon_1(t)) \mathcal{C}_2 + \mathcal{C}_3(\Upsilon_2^{\diamond}(t) - \Upsilon_1^{\diamond}(t)) \mathcal{C}_4] \right. \\ &\quad \left. \times g(\varphi(\varrho, qt)) + \mathcal{U}(\varrho, t) \right)^H e(\varrho, t) \Big] d\varrho. \end{aligned} \quad (11)$$

Through synergistic application of Green's formula and Lemma 1, we establish

$$\begin{aligned} &\int_{\Omega} e^H(\varrho, t) \mathbb{K}\{e(\varrho, t)\} d\varrho \\ &= \sum_{\kappa=1}^n \int_{\Omega} \overline{e_{\kappa}(\varrho, t)} \sum_{s=1}^S \frac{\partial}{\partial \varrho_s} \left(\beta_{\kappa s} \frac{\partial e_{\kappa}(\varrho, t)}{\partial \varrho_s} \right) d\varrho \\ &= \sum_{\kappa=1}^n \int_{\Omega} \overline{e_{\kappa}(\varrho, t)} \nabla \cdot \left(\beta_{\kappa s} \frac{\partial e_{\kappa}(\varrho, t)}{\partial \varrho_s} \right)_{s=1}^S d\varrho \\ &= \sum_{\kappa=1}^n \int_{\Omega} \left[\nabla \cdot \left(\overline{e_{\kappa}(\varrho, t)} \beta_{\kappa s} \frac{\partial e_{\kappa}(\varrho, t)}{\partial \varrho_s} \right)_{s=1}^S \right. \end{aligned}$$

$$\begin{aligned} &\quad \left. - \left(\beta_{\kappa s} \frac{\partial e_{\kappa}(\varrho, t)}{\partial \varrho_s} \right)_{s=1}^S \cdot \nabla \overline{e_{\kappa}(\varrho, t)} \right] d\varrho \\ &\leq \sum_{\kappa=1}^n \int_{\Omega} \left(\overline{e_{\kappa}(\varrho, t)} \beta_{\kappa s} \frac{\partial e_{\kappa}(\varrho, t)}{\partial \varrho_s} \right)_{s=1}^S \cdot \nabla d\varrho \\ &\quad - \sum_{\kappa=1}^n \sum_{s=1}^S \int_{\Omega} \beta_{\kappa s} \left| \frac{\partial e_{\kappa}(\varrho, t)}{\partial \varrho_s} \right|^2 d\varrho \\ &= - \sum_{\kappa=1}^n \sum_{s=1}^S \int_{\Omega} \beta_{\kappa s} \left| \frac{\partial e_{\kappa}(\varrho, t)}{\partial \varrho_s} \right|^2 d\varrho \end{aligned}$$

with

$$\int_{\Omega} \mathbb{K}^H\{e(\varrho, t)\} e(\varrho, t) d\varrho = - \sum_{\kappa=1}^n \sum_{s=1}^S \int_{\Omega} \beta_{\kappa s} \left| \frac{\partial e_{\kappa}(\varrho, t)}{\partial \varrho_s} \right|^2 d\varrho$$

then

$$\begin{aligned} &\int_{\Omega} [e^H(\varrho, t) \mathbb{K}\{e(\varrho, t)\} + \mathbb{K}^H\{e(\varrho, t)\} e(\varrho, t)] d\varrho \\ &= - \sum_{\kappa=1}^n \sum_{s=1}^S \int_{\Omega} 2\beta_{\kappa s} \left(\frac{\partial e_{\kappa}(\varrho, t)}{\partial \varrho_s} \right)^2 d\varrho \\ &\leq - \sum_{\kappa=1}^n \sum_{s=1}^S \int_{\Omega} \frac{2\beta_{\kappa s}}{\theta_s^2} |e_{\kappa}(\varrho, t)|^2 d\varrho. \end{aligned} \quad (12)$$

Applying (A₂) yields

$$\begin{aligned} &\int_{\Omega} [e^H(\varrho, t) B^{\Delta} f(e(\varrho, t)) + (B^{\Delta} f(e(\varrho, t)))^H e(\varrho, t)] d\varrho \\ &= \int_{\Omega} 2\text{Re}(e^H(\varrho, t) B^{\Delta} f(e(\varrho, t))) d\varrho \\ &\leq 2\|e(\varrho, t)\| \|B^{\Delta}\| \|f(e(\varrho, t))\| \\ &\leq 2F\|e(\varrho, t)\| \|B^{\Delta}\| \|e(\varrho, t)\| \\ &\leq \|e(\varrho, t)\|^2 \|B^{\Delta}\|^2 + \|e(\varrho, t)\|^2 F^2 \end{aligned} \quad (13)$$

and

$$\begin{aligned} &\int_{\Omega} [e^H(\varrho, t) \mathcal{B}_1(\Theta_2(t) - \Theta_1(t)) \mathcal{B}_2 f(\varphi(\varrho, t)) \\ &\quad + (\mathcal{B}_1(\Theta_2(t) - \Theta_1(t)) \mathcal{B}_2 f(\varphi(\varrho, t)))^H e(\varrho, t)] d\varrho \\ &\leq 2\|e(\varrho, t)\| \|\mathcal{B}_1(\Theta_2(t) - \Theta_1(t)) \mathcal{B}_2\| \mathcal{L}_1 \\ &\leq 4\|e(\varrho, t)\| \|\mathcal{B}_1 \mathcal{B}_2\| \mathcal{L}_1. \end{aligned} \quad (14)$$

Applying the computational framework from (13) and (14) to the analogous terms in (11), and substituting (12) into (10), we obtain

$$\begin{aligned} \frac{\partial \mathcal{V}_1(\varrho, t)}{\partial t} &\leq \sum_{p=1}^P \zeta_p(\mu(t)) \left[\delta (1+t)^{\delta-1} \|e(\varrho, t)\|^2 \right. \\ &\quad \left. + (1+t)^{\delta} \left\{ \int_{\Omega} \left[- \sum_{\kappa=1}^n \sum_{s=1}^S \frac{\beta_{\kappa s}}{\theta_s^2} |e_{\kappa}(\varrho, t)|^2 \right. \right. \right. \\ &\quad - 2e^H(\varrho, t) D^{[p]} e(\varrho, t) + e^H(\varrho, t) \mathcal{U}(\varrho, t) \\ &\quad + (\mathcal{U}(\varrho, t))^H e(\varrho, t) \Big] d\varrho + \|e(\varrho, t)\|^2 (\|B^{\Delta}\|^2 \\ &\quad + \|B^*\|^2 + \|\mathcal{B}_1 \mathcal{B}_2\|^2 + \|\mathcal{B}_3 \mathcal{B}_4\|^2 + \Delta b + 5F^2 \\ &\quad \left. \left. + \|C^{\Delta}\|^2 + \|C^*\|^2 + \|\mathcal{C}_1 \mathcal{C}_2\|^2 + \|\mathcal{C}_3 \mathcal{C}_4\|^2 \right) \right] \end{aligned}$$

$$+ \Delta c) + 5G^2 \|e(\varrho, qt)\|^2 + 4\|e(\varrho, t)\| (\|B_1 B_2\| \mathcal{L}_1 + \|B_3 B_4\| \mathcal{L}_1 + \|C_1 C_2\| \mathcal{L}_2 + \|C_3 C_4\| \mathcal{L}_2) \}. \quad (15)$$

Substituting controller (8) yields

$$\begin{aligned} e^H(\varrho, t) \mathcal{W}(\varrho, t) + (\mathcal{W}(\varrho, t))^H e(\varrho, t) \\ = -e^H(\varrho, t) \left(\mathcal{W}(\varrho, t) + \mathcal{M}(\varrho, t) \|e(\varrho, t)\|^{-1} \right) e(\varrho, t_r) \\ - e^H(\varrho, t_r) \left(\mathcal{W}^H(\varrho, t) + \mathcal{M}^H(\varrho, t) \|e(\varrho, t)\|^{-1} \right) e(\varrho, t) \\ = - \sum_{k=1}^n \left[\overline{e_k(\varrho, t)} \left(\mathcal{W}_k(\varrho, t) + \mathcal{M}_k(\varrho, t) \|e(\varrho, t)\|^{-1} \right) \right. \\ \times e_k(\varrho, t_r) + e_k(\varrho, t) \left(\overline{\mathcal{W}_k(\varrho, t)} + \overline{\mathcal{M}_k(\varrho, t)} \right) \\ \times \left. \|e(\varrho, t)\|^{-1} \overline{e_k(\varrho, t_r)} \right]. \quad (16) \end{aligned}$$

Combining the constraint $q_i \in (0, 1)$, the definition of $\hat{q}$, and $\delta > 0$, we derive

$$(1 + q_i^{-1}t)^\delta \leq q_i^{-\delta}(1+t)^\delta \leq \hat{q}^{-\delta}(1+t)^\delta. \quad (17)$$

Employing the same operator for $\mathcal{V}_2(\varrho, t)$ and $\mathcal{V}_3(\varrho, t)$ derivatives, in conjunction with (17), leads to

$$\begin{aligned} \frac{\partial \mathcal{V}_2(\varrho, t)}{\partial t} &= \int_{\Omega} 5G^2 \sum_{i=1}^n [q_i^{-1} (1 + q_i^{-1}t)^\delta |e_i(\varrho, t)|^2 \\ &\quad - (1+t)^\delta |e_i(\varrho, q_i t)|^2] d\varrho \\ &\leq 5G^2 \left[\hat{q}^{-(\delta+1)} (1+t)^\delta \|e(\varrho, t)\|^2 \right. \\ &\quad \left. - (1+t)^\delta \|e(\varrho, qt)\|^2 \right] \\ &\leq 5G^2 \sum_{p=1}^P \zeta_p(\mu(t)) (1+t)^\delta \\ &\quad \times \left[\hat{q}^{-(\delta+1)} \|e(\varrho, t)\|^2 - \|e(\varrho, qt)\|^2 \right] \quad (18) \end{aligned}$$

$$\begin{aligned} \frac{\partial \mathcal{V}_3(\varrho, t)}{\partial t} &= \int_{\Omega} \left\{ \sum_{k=1}^n \sum_{p=1}^P \zeta_p(\mu(t)) (1+t)^\delta \left[\overline{e_k(\varrho, t)} (\mathcal{W}_k(\varrho, t) \right. \right. \\ &\quad \left. \left. + \mathcal{M}_k(\varrho, t) \|e(\varrho, t)\|^{-1}) e_k(\varrho, t_r) + \overline{e_k(\varrho, t_r)} \right. \right. \\ &\quad \times \left(\overline{\mathcal{W}_k(\varrho, t)} + \overline{\mathcal{M}_k(\varrho, t)} \|e(\varrho, t)\|^{-1} \right) e_k(\varrho, t) \\ &\quad - \mathcal{W}_k e_k(\varrho, t) e_k(\varrho, t_r) - \overline{\mathcal{W}_k} e_k(\varrho, t) \overline{e_k(\varrho, t_r)} \\ &\quad - \mathcal{M}_k \overline{e_k(\varrho, t)} \|e(\varrho, t)\|^{-1} e_k(\varrho, t_r) \\ &\quad \left. \left. - \overline{\mathcal{M}_k} e_k(\varrho, t) \|e(\varrho, t)\|^{-1} \overline{e_k(\varrho, t_r)} \right] \right\} d\varrho \quad (19) \end{aligned}$$

in which

$$\begin{aligned} \sum_{k=1}^n &\left[-\mathcal{W}_k \overline{e_k(\varrho, t)} e_k(\varrho, t_r) - \overline{\mathcal{W}_k} e_k(\varrho, t) \overline{e_k(\varrho, t_r)} \right. \\ &\quad \left. - \mathcal{M}_k \overline{e_k(\varrho, t)} \|e(\varrho, t)\|^{-1} e_k(\varrho, t_r) \right. \\ &\quad \left. - \overline{\mathcal{M}_k} e_k(\varrho, t) \|e(\varrho, t)\|^{-1} \overline{e_k(\varrho, t_r)} \right] \\ &= -\mathcal{W}_k |e_k(\varrho, t)|^2 + \mathcal{W}_k \overline{e_k(\varrho, t)} (e_k(\varrho, t) - e_k(\varrho, t_r)) \\ &\quad - \overline{\mathcal{W}_k} |e_k(\varrho, t)|^2 + \overline{\mathcal{W}_k} e_k(\varrho, t) (\overline{e_k(\varrho, t)} - \overline{e_k(\varrho, t_r)}) \\ &\quad - \mathcal{M}_k |e_k(\varrho, t)|^2 \|e(\varrho, t)\|^{-1} + \mathcal{M}_k \overline{e_k(\varrho, t)} \|e(\varrho, t)\|^{-1} \\ &\quad \times (e_k(\varrho, t) - e_k(\varrho, t_r)) - \overline{\mathcal{M}_k} |e_k(\varrho, t)|^2 \|e(\varrho, t)\|^{-1} \end{aligned}$$

$$+ \overline{\mathcal{M}_k} e_k(\varrho, t) \|e(\varrho, t)\|^{-1} (\overline{e_k(\varrho, t)} - \overline{e_k(\varrho, t_r)}).$$

According to the event-triggered mechanism (7), we have

$$\begin{aligned} \int_{\Omega} \sum_{k=1}^n &\left[\mathcal{W}_k \overline{e_k(\varrho, t)} (e_k(\varrho, t) - e_k(\varrho, t_r)) \right. \\ &\quad \left. + \overline{\mathcal{W}_k} e_k(\varrho, t) (\overline{e_k(\varrho, t)} - \overline{e_k(\varrho, t_r)}) \right] d\varrho \\ &\leq 2\|\mathcal{W}\| \|e(\varrho, t)\| \|e(\varrho, t) - e(\varrho, t_r)\| \\ &\leq 2\lambda \|\mathcal{W}\| \|e(\varrho, t)\|^2 \quad (20) \end{aligned}$$

$$\begin{aligned} \int_{\Omega} \sum_{k=1}^n &\left[\mathcal{M}_k \overline{e_k(\varrho, t)} \|e(\varrho, t)\|^{-1} (e_k(\varrho, t) - e_k(\varrho, t_r)) \right. \\ &\quad \left. + \overline{\mathcal{M}_k} e_k(\varrho, t) \|e(\varrho, t)\|^{-1} (\overline{e_k(\varrho, t)} - \overline{e_k(\varrho, t_r)}) \right] d\varrho \\ &\leq 2\|\mathcal{M}\| \|e(\varrho, t) - e(\varrho, t_r)\| \leq 2\lambda \|\mathcal{M}\| \|e(\varrho, t)\| \quad (21) \end{aligned}$$

$$\begin{aligned} \int_{\Omega} \sum_{k=1}^n &-(\mathcal{M}_k + \overline{\mathcal{M}_k}) |e_k(\varrho, t)|^2 \|e(\varrho, t)\|^{-1} d\varrho \\ &\leq -\mathcal{M}_v \|e(\varrho, t)\| \quad (22) \end{aligned}$$

and

$$\int_{\Omega} \sum_{k=1}^n -(\mathcal{W}_k + \overline{\mathcal{W}_k}) |e_k(\varrho, t)|^2 d\varrho \leq -\mathcal{W}_v \|e(\varrho, t)\|^2 \quad (23)$$

with $\mathcal{W}_v = \min_k \{\mathcal{W}_k + \overline{\mathcal{W}_k}\}$, $\mathcal{M}_v = \min_k \{\mathcal{M}_k + \overline{\mathcal{M}_k}\}$, $\mathcal{W} = \text{diag}(\mathcal{W}_k)$, $\mathcal{M} = \text{diag}(\mathcal{M}_k)$.

Substituting (16) into (15), and (20)–(23) into (19), then combining with (12) and (18), we obtain

$$\begin{aligned} \frac{\partial \mathcal{V}(\varrho, t)}{\partial t} &\leq \sum_{p=1}^P \zeta_p(\mu(t)) (1+t)^\delta \\ &\quad \times \left[\int_{\Omega} \left[- \sum_{k=1}^n \sum_{s=1}^S \frac{\beta_{ks}}{\theta_s^2} |e_k(\varrho, t)|^2 \right. \right. \\ &\quad \left. \left. - 2e^H(\varrho, t) D^{[p]} e(\varrho, t) \right] d\varrho + \|e(\varrho, t)\|^2 \left(\delta + \|B^\Delta\|^2 \right. \right. \\ &\quad \left. \left. + \|B^*\|^2 + \|B_1 B_2\|^2 + \|B_3 B_4\|^2 + \Delta b + 5F^2 \right. \right. \\ &\quad \left. \left. + \|C^\Delta\|^2 + \|C^*\|^2 + \|C_1 C_2\|^2 + \|C_3 C_4\|^2 + \Delta c \right. \right. \\ &\quad \left. \left. + 5G^2 \hat{q}^{-(\delta+1)} + 2\lambda \|\mathcal{W}\| - \mathcal{W}_v \right) \right. \\ &\quad \left. \left. + 4\|e(\varrho, t)\| (\|B_1 B_2\| \mathcal{L}_1 + \|B_3 B_4\| \mathcal{L}_1 \right. \right. \\ &\quad \left. \left. + \|C_1 C_2\| \mathcal{L}_2 + \|C_3 C_4\| \mathcal{L}_2 + \frac{1}{2}\lambda \|\mathcal{M}\| - \frac{1}{4}\mathcal{M}_v) \right] \right]. \end{aligned}$$

Choosing

$$\begin{aligned} \mathcal{W}_v &= \delta + \|B^\Delta\|^2 + \|B^*\|^2 + \|B_1 B_2\|^2 + \|B_3 B_4\|^2 \\ &\quad + \Delta b + 5F^2 + \|C^\Delta\|^2 + \|C^*\|^2 + \|C_1 C_2\|^2 \\ &\quad + \|C_3 C_4\|^2 + \Delta c + 5G^2 \hat{q}^{-(\delta+1)} + 2\lambda \|\mathcal{W}\| \end{aligned}$$

and

$$\begin{aligned} \mathcal{M}_v &= 4\|B_1 B_2\| \mathcal{L}_1 + 4\|B_3 B_4\| \mathcal{L}_1 \\ &\quad + 4\|C_1 C_2\| \mathcal{L}_2 + 4\|C_3 C_4\| \mathcal{L}_2 + 2\lambda \|\mathcal{M}\|. \end{aligned}$$

We obtain

$$\frac{\partial \mathcal{V}(\varrho, t)}{\partial t} \leq \sum_{p=1}^P \zeta_p(\mu(t)) (1+t)^\delta \int_{\Omega} \left[- \sum_{k=1}^n \sum_{s=1}^S \frac{\beta_{ks}}{\theta_s^2} \right.$$

$$\times |e_k(\varrho, t)|^2 - 2e^H(\varrho, t) D^{[p]} e(\varrho, t) \Big] d\varrho \leq 0.$$

Then

$$\mathcal{V}(\varrho, t) \leq \mathcal{V}(\varrho, t_0) \quad (24)$$

with

$$\begin{aligned} \mathcal{V}(\varrho, t_0) &= \int_{\Omega} [(1+t_0)^{\delta} e^H(\varrho, t_0) e(\varrho, t_0) + \delta + 5G^2 \\ &\quad \times \sum_{i=1}^n \int_{q_i t_0}^{t_0} (1+q_i^{-1} \varsigma)^{\delta} q_i^{-1} |e_i(\varrho, \varsigma)|^2 d\varsigma] d\varrho \\ &\leq (1+t_0)^{\delta} \|e(\varrho, t_0)\|^2 + \int_{\Omega} \delta d\varrho + 5G^2 \\ &\quad \times \int_{\Omega} \sum_{i=1}^n q_i^{-(\delta+1)} (1+t_0)^{\delta+1} \sup_{\mathcal{U}} \{|e_i(\varrho, t)|^2\} d\varrho \\ &\leq (1+t_0)^{\delta} \max_{1 \leq k, i \leq n} \left\{ 1 + 5G^2 (1+t_0) \sum_{i=1}^n q_i^{-(\delta+1)} \right\} \\ &\quad \times \int_{\Omega} \sum_{k=1}^n \sup_{\mathcal{U}} \{|e_k(\varrho, t)|^2\} d\varrho + \int_{\Omega} \delta d\varrho \\ &\leq \aleph \|\widetilde{\omega} - \omega\|_{\hat{q}}^2 (1+t_0)^{\delta} \end{aligned} \quad (25)$$

where

$$\begin{aligned} \delta &= (1+t_0)^{-\delta} \sum_{k=1}^n [\xi_{1k}^{-1} (\overline{\mathcal{W}_k(\varrho, t_0)} - \mathcal{W}_k(\varrho, t_0) - \mathcal{W}_k) \\ &\quad + \xi_{2k}^{-1} (\overline{\mathcal{M}_k(\varrho, t_0)} - \mathcal{M}_k(\varrho, t_0) - \mathcal{M}_k)] \end{aligned}$$

and

$$\begin{aligned} \aleph &= 1 + \max_{1 \leq k, i \leq n} \left\{ 1 + 5G^2 (1+t_0) \sum_{i=1}^n q_i^{-(\delta+1)} \right\} \\ &\quad + \delta \left(\|\widetilde{\omega} - \omega\|_{\hat{q}}^2 \right)^{-1} \geq 1. \end{aligned}$$

Drawing upon (9), (23), and (24) collectively, it can be established that

$$\mathcal{V}_1(\varrho, t) \leq \mathcal{V}(\varrho, t) \leq \mathcal{V}(\varrho, t_0).$$

Thus

$$(1+t)^{\delta} \|e(\varrho, t)\|^2 \leq \aleph \|\widetilde{\omega} - \omega\|_{\hat{q}}^2 (1+t_0)^{\delta}$$

i.e.,

$$\|e(\varrho, t)\|^2 \leq \aleph \|\widetilde{\omega} - \omega\|_{\hat{q}}^2 \left(\frac{1+t}{1+t_0} \right)^{-\delta}.$$

Per Definition 1, GPS between systems (4) and (5) is attainable. ■

Zeno behavior, defined by infinite triggers in finite time, can cause system failure. It directly contradicts the core goal of resource conservation in event-triggered control and results in maximum resource consumption. Thus, guaranteeing a strictly positive minimum inter-event time $t_{r+1} - t_r > 0$ is imperative. We now examine whether the event-triggered controller (8) can avoid Zeno behavior.

Theorem 2: Zeno behavior is excluded in the adaptive event-triggered controller (8).

Proof: For $t \in [t_r, t_{r+1})$, let $\widetilde{e}(\varrho, t) = e(\varrho, t_r) - e(\varrho, t)$, one has

$$\frac{\partial \|\widetilde{e}(\varrho, t)\|}{\partial t} \leq \left\| \frac{\partial \widetilde{e}(\varrho, t)}{\partial t} \right\| = \left\| \frac{\partial e(\varrho, t)}{\partial t} \right\|$$

$$\begin{aligned} &\leq \|\mathbb{K}\{e(\varrho, t)\}\| + \|e(\varrho, t)\| \left(\|D^{[p]}\| + F(\|B^{\Delta}\| \right. \\ &\quad \left. + \|B^{\circ}\| + \|\mathcal{B}_1 \mathcal{B}_2\| + \|\mathcal{B}_3 \mathcal{B}_4\| + \Delta b) + 2\mathcal{L}_1 \right. \\ &\quad \left. \times (\|\mathcal{B}_1 \mathcal{B}_2\| + \|\mathcal{B}_3 \mathcal{B}_4\|) + \|e(\varrho, qt)\| G(\|C^{\Delta}\| \right. \\ &\quad \left. + \|C^{\circ}\| + \|\mathcal{C}_1 \mathcal{C}_2\| + \|\mathcal{C}_3 \mathcal{C}_4\| + \Delta c) \right. \\ &\quad \left. + 2\mathcal{L}_2 (\|\mathcal{C}_1 \mathcal{C}_2\| + \|\mathcal{C}_3 \mathcal{C}_4\|) + \|e(\varrho, t_r)\| \right. \\ &\quad \left. \times (\|\mathcal{W}(\varrho, t)\| + \|e(\varrho, t)\|^{-1} \|\mathcal{M}(\varrho, t)\|) \right). \end{aligned} \quad (26)$$

Based on Theorem 1, for $t \in [t_r, t_{r+1})$, the conditions $\mathcal{V}_1(\varrho, t) \leq \mathcal{V}(\varrho, t)$ and $(\partial \mathcal{V}(\varrho, t))/(\partial t) \leq 0$ hold. This leads to $\mathcal{V}_1(\varrho, t) \leq \mathcal{V}(\varrho, t) \leq \mathcal{V}(\varrho, t_r) \leq \mathcal{V}(\varrho, \hat{q}t_r)$. Furthermore, since $\|e(\varrho, t)\|^2 \leq (1+t)^{\delta} \|e(\varrho, t)\|^2 = \mathcal{V}_1(\varrho, t)$, we derive $\|e(\varrho, t)\| \leq (\mathcal{V}(\varrho, t_r))^{1/2}$ and $\|e(\varrho, qt)\| \leq (\mathcal{V}(\varrho, \hat{q}t_r))^{1/2}$.

Let

$$\begin{aligned} \wp &= \max_{t \in [t_r, t_{r+1})} \left\{ \|\mathcal{W}(\varrho, t)\| + \|e(\varrho, t)\|^{-1} \|\mathcal{M}(\varrho, t)\| \right\} \\ \alpha &= \|D^{[p]}\| + F(\|B^{\Delta}\| + \|B^{\circ}\| + \|\mathcal{B}_1 \mathcal{B}_2\| + \|\mathcal{B}_3 \mathcal{B}_4\| + \Delta b) \\ &\quad + G(\|C^{\Delta}\| + \|C^{\circ}\| + \|\mathcal{C}_1 \mathcal{C}_2\| + \|\mathcal{C}_3 \mathcal{C}_4\| + \Delta c) \end{aligned}$$

and

$$\begin{aligned} \gamma &= \mathcal{L}_1 (\|\mathcal{B}_1 \mathcal{B}_2\| + \|\mathcal{B}_3 \mathcal{B}_4\|) + \mathcal{L}_2 (\|\mathcal{C}_1 \mathcal{C}_2\| + \|\mathcal{C}_3 \mathcal{C}_4\|) \\ &\quad + \max_{t \in [t_r, t_{r+1})} \left\{ \|\mathbb{K}\{e(\varrho, t)\}\| \right\}. \end{aligned}$$

Obviously

$$\frac{\partial \|\widetilde{e}(\varrho, t)\|}{\partial t} \leq \alpha \sqrt{\mathcal{V}(\varrho, \hat{q}t_r)} + \gamma + \wp \|e(\varrho, t_r)\|.$$

Since $\widetilde{e}(\varrho, t_r) = 0$, integrating the preceding formula from t_r to t yields $\|\widetilde{e}(\varrho, t)\| \leq (\alpha(\mathcal{V}(\varrho, \hat{q}t_r))^{1/2} + \gamma + \wp \|e(\varrho, t_r)\|)(t - t_r)$.

We analyze the minimal time interval through three stages.

Step I: Error Bound Derivation:

$\|\widetilde{e}(\varrho, t)\| \leq \lambda \|e(\varrho, t)\| \leq \lambda \|\widetilde{e}(\varrho, t)\| + \lambda \|e(\varrho, t_r)\|$, which yields $\|\widetilde{e}(\varrho, t)\| \leq \lambda/(1 - \lambda) \|e(\varrho, t_r)\|$.

Step II: Triggering Condition at t_{r+1} :

$\|\widetilde{e}(\varrho, t_{r+1})\| > \lambda/(1 - \lambda) \|e(\varrho, t_r)\|$ and $\|\widetilde{e}(\varrho, t_{r+1})\| \leq (\alpha(\mathcal{V}(\varrho, \hat{q}t_r))^{1/2} + \gamma + \wp \|e(\varrho, t_r)\|)(t_{r+1} - t_r)$.

Step III: Time Interval Lower Bound:

Synthesizing the requirements in both steps leads to $\|e(\varrho, t_r)\| < 1 - \lambda/(\lambda(\alpha(\mathcal{V}(\varrho, \hat{q}t_r))^{1/2} + \gamma + \wp \|e(\varrho, t_r)\|)(t_{r+1} - t_r)$, therefore $t_{r+1} - t_r > (\lambda \|e(\varrho, t_r)\|)/((1 - \lambda)(\alpha(\mathcal{V}(\varrho, \hat{q}t_r))^{1/2} + \gamma + \wp \|e(\varrho, t_r)\|)) > 0$. Consequently, Zeno phenomena are excluded. ■

When $\delta = 0$, GPS degenerates to GAS, namely, the following.

Corollary 1: Under (A_1) , (A_2) , and

$$\mathcal{U}(\varrho, t) = -\mathcal{W}(\varrho, t) e(\varrho, t_r) - \mathcal{M}(\varrho, t) \frac{e(\varrho, t_r)}{\|e(\varrho, t)\|} \quad (27)$$

with $t \in [t_r, t_{r+1})$, and the remaining parameters are designed consistently with those in the adaptive event-triggered controller (8), then GAS between systems (4) and (5) is attainable.

Corollary 2: Under (A_1) , (A_2) , and

$$\mathcal{U}(\varrho, t) = -\mathcal{W}^* e(\varrho, t_r) - \mathcal{M}^* \frac{e(\varrho, t_r)}{\|e(\varrho, t)\|} \quad (28)$$

with $t \in [t_r, t_{r+1})$, $\mathcal{W}^* = \text{diag}(\mathcal{W}_k^*)$, $\mathcal{M}^* = \text{diag}(\mathcal{M}_k^*) \in \mathbb{C}^{n \times n}$, $\mathcal{W}_v^* = \min_k \{\mathcal{W}_k^* + \overline{\mathcal{W}_k^*}\}$, $\mathcal{M}_v^* = \min_k \{\mathcal{M}_k^* + \overline{\mathcal{M}_k^*}\}$, $\mathcal{W}^* = \text{diag}(\mathcal{W}_k^*)$, $\mathcal{M}^* = \text{diag}(\mathcal{M}_k^*)$, such that

$$\begin{aligned} \square_1 = & \delta + \|\mathcal{B}^\Delta\|^2 + \|\mathcal{B}^\bullet\|^2 + \|\mathcal{B}_1\mathcal{B}_2\|^2 + \|\mathcal{B}_3\mathcal{B}_4\|^2 + \Delta b \\ & + 5F^2 + \|\mathcal{C}^\Delta\|^2 + \|\mathcal{C}^\bullet\|^2 + \|\mathcal{C}_1\mathcal{C}_2\|^2 + \|\mathcal{C}_3\mathcal{C}_4\|^2 \\ & + \Delta c + 5G^2\hat{q}^{-(\delta+1)} + 2\lambda\|\mathcal{W}^*\| - \mathcal{W}_v^* \leq 0 \end{aligned} \quad (29)$$

$$\begin{aligned} \square_2 = & 4\|\mathcal{B}_1\mathcal{B}_2\|\mathcal{L}_1 + 4\|\mathcal{B}_3\mathcal{B}_4\|\mathcal{L}_1 + 4\|\mathcal{C}_1\mathcal{C}_2\|\mathcal{L}_2 \\ & + 4\|\mathcal{C}_3\mathcal{C}_4\|\mathcal{L}_2 + 2\lambda\|\mathcal{M}^*\| - \mathcal{M}_v^* \leq 0 \end{aligned} \quad (30)$$

then GPS between systems (4) and (5) is attainable.

Corollary 3: Under (A_1) , (A_2) , and

$$\mathcal{U}(\varrho, t) = -\mathcal{W}^*e(\varrho, t_r) - \mathcal{M}^* \frac{e(\varrho, t_r)}{\|e(\varrho, t)\|}$$

with $t \in [t_r, t_{r+1})$, $\mathcal{W}^* = \text{diag}(\mathcal{W}_k^*)$, $\mathcal{M}^* = \text{diag}(\mathcal{M}_k^*) \in \mathbb{C}^{n \times n}$, $\mathcal{W}_v^* = \min_k \{\mathcal{W}_k^* + \overline{\mathcal{W}_k^*}\}$, $\mathcal{M}_v^* = \min_k \{\mathcal{M}_k^* + \overline{\mathcal{M}_k^*}\}$, $\mathcal{W}^* = \text{diag}(\mathcal{W}_k^*)$, $\mathcal{M}^* = \text{diag}(\mathcal{M}_k^*)$, such that (29) holds and

$$\begin{aligned} \square_1 = & \|\mathcal{B}^\Delta\|^2 + \|\mathcal{B}^\bullet\|^2 + \|\mathcal{B}_1\mathcal{B}_2\|^2 + \|\mathcal{B}_3\mathcal{B}_4\|^2 + \Delta b \\ & + 5F^2 + \|\mathcal{C}^\Delta\|^2 + \|\mathcal{C}^\bullet\|^2 + \|\mathcal{C}_1\mathcal{C}_2\|^2 + \|\mathcal{C}_3\mathcal{C}_4\|^2 \\ & + \Delta c + 5G^2\hat{q}^{-1} + 2\lambda\|\mathcal{W}^*\| - \mathcal{W}_v^* \leq 0 \end{aligned}$$

then GAS between systems (4) and (5) is attainable.

Remark 3: This article develops an adaptive event-triggered controller that merges the resource efficiency of event-triggering with the self-regulating capability of adaptive control. This combination yields a structurally simpler design than those in [4], [5], and [19], and simplifies the assurance of its theoretical conditions. Consequently, our main result requires only assumptions (A_1) and (A_2) , making it more verifiable and enhancing its practical implementability.

Remark 4: The condition remains valid when $\lambda = q_i$ ($i \in \mathcal{N}$), as both the event-triggering parameter λ and the proportional delay factor q_i , share the same interval $(0, 1)$.

Remark 5: This work distinguishes itself by considering proportional delays, in contrast to the delay-free model in [41] and the constant-delay model in [42]. The proposed solution is further enhanced by an adaptive event-triggered controller, which achieves both energy savings and a relaxation of the conditions required for theoretical verification.

Remark 6: An additional limitation lies in the restriction to 1-D spatial domains [9], [13], despite the fact that most practical reaction–diffusion processes take place in two or three dimensions. This choice is common in foundational studies due to the mathematical simplicity of the Laplacian operator, easier treatment of boundary conditions, and more straightforward stability analysis in one dimension. Nevertheless, such a simplification inherently constrains the model's applicability to real-world systems.

IV. SIMULATION EXAMPLES

Example 1: The numerical simulation is based on the 2-D systems (4) and (5). The parameters are set as: $\beta_{1s} = \beta_{2s} = \beta_{3s} = 0.1$, and the activation functions are $f_k(\varphi) = g_k(\varphi) = \tanh(\text{Re}(\varphi)) + \tanh(\text{Im}(\varphi))i$, for $k = 1, 2$, with $L = G = 1$ and $\mathcal{L}_1 = \mathcal{L}_2 = \sqrt{2}$.

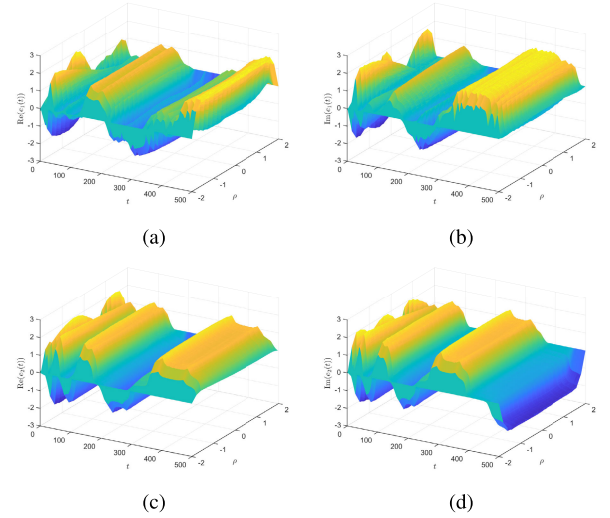


Fig. 3. SRCs of the real and imaginary parts for uncontrolled system (6). (a) $\text{Re}(e_1(t))$, (b) $\text{Im}(e_1(t))$, (c) $\text{Re}(e_2(t))$, and (d) $\text{Im}(e_2(t))$.

The parameter uncertainties are given by: $\Delta B(\varrho, t) = 10^{-3} \times \begin{pmatrix} 2 \cos(2t) + 3 \sin(2t)i & \tanh(t) + 2 \sin(4t)i \\ 1.5 \tanh(3t) + 4 \sin(t)i & 2 \cos(2t) + \sin(4t)i \end{pmatrix}$, $\Delta C(\varrho, t) = 10^{-3} \begin{pmatrix} \sin(2t) + 3 \tanh(3t)i & \tanh(t) + \sin(3t)i \\ 2 \sin(2t) + 3.5 \cos(t)i & 3 \cos(t) + \tanh(3t)i \end{pmatrix}$.

The spatial domain is $\varrho \in [-2, 2]$, with proportional delay factors $q_1 = 0.85$, $q_2 = 0.75$, $\hat{q} = 0.75$. $\delta = 1.4$. It can be calculated that $\Delta b = 0.002$ and $\Delta c = 0.0047$. The memristive connection values evolve as follows:

$$\begin{aligned} b_{11}(\cdot) &= \begin{cases} 3.5 + 1.8i, & \dagger_{11} \\ -3 - 1.5i, & \dagger_{12} \end{cases} & b_{12}(\cdot) &= \begin{cases} 3 + 1.5i, & \dagger_{11} \\ -2.8 - 1.2i, & \dagger_{12} \end{cases} \\ b_{21}(\cdot) &= \begin{cases} 2.8 + 1.2i, & \dagger_{21} \\ -2.5 - i, & \dagger_{22} \end{cases} & b_{22}(\cdot) &= \begin{cases} 2.6 + 1.4i, & \dagger_{21} \\ -2.3 - 1.4i, & \dagger_{22} \end{cases} \\ c_{11}(\cdot) &= \begin{cases} 1.8 + 0.9i, & \dagger_{11} \\ -1.5 - 0.75i, & \dagger_{12} \end{cases} & c_{12}(\cdot) &= \begin{cases} 1.5 + 0.75i, & \dagger_{11} \\ -1.2 - 0.6i, & \dagger_{12} \end{cases} \\ c_{21}(\cdot) &= \begin{cases} 1.2 + 0.6i, & \dagger_{21} \\ -0.9 - 0.45i, & \dagger_{22} \end{cases} & c_{22}(\cdot) &= \begin{cases} 1.4 + 0.55i, & \dagger_{21} \\ -1.7 - 0.35i, & \dagger_{22} \end{cases} \end{aligned}$$

with $\mathcal{T}_k = 1$, $\kappa = 1, 2$.

The system initial conditions are selected as follows: $\varphi_1(\varrho, t) = 0.1 \cos(\pi\varrho)(1 + 0.5 \cos(2\pi\varrho)) + 0.05 \sin(\pi\varrho)(1 + 0.3 \cos(3\pi\varrho))i$, $\varphi_2(\varrho, t) = 0.08 \sin(\pi\varrho)(1 + 0.4 \cos(\pi\varrho)) + 0.06 \cos(\pi\varrho)(1 + 0.2 \sin(2\pi\varrho))i$, $\psi_1(\varrho, t) = 0.15 \sin(\pi\varrho)(1 + 0.4 \cos(2\pi\varrho)) + 0.09 \cos(\pi\varrho)(1 + 0.3 \sin(2\pi\varrho))i$, and $\psi_2(\varrho, t) = 0.12 \cos(\pi\varrho)(1 + 0.3 \sin(2\pi\varrho)) + 0.07 \sin(\pi\varrho)(1 + 0.5 \cos(\pi\varrho))i$.

Considering the case with two T–S fuzzy rules ($p = 2$), the computational logic of the drive system is defined as follows, and similarly for the response system.

Donote $\tau_{11} = \chi_1^{(1)}(\mu_1(t)) = \exp(-2(|\varphi_1(\varrho, t)| - 0.5)^2)$, $\tau_{12} = \chi_2^{(1)}(\mu_2(t)) = \exp(-2(|\varphi_2(\varrho, t)| + 0.5)^2)$, $\tau_{21} = \chi_1^{(2)}(\mu_1(t)) = \exp(-2(|\varphi_1(\varrho, t)| - 1)^2)$, $\tau_{22} = \chi_2^{(2)}(\mu_2(t)) = \exp(-2(|\varphi_2(\varrho, t)| + 1)^2)$, $\zeta_1(\mu(t)) = (\tau_{11}\tau_{12})/(\tau_{11}\tau_{12} + \tau_{21}\tau_{22})$, and $\zeta_2(\mu(t)) = (\tau_{21}\tau_{22})/(\tau_{11}\tau_{12} + \tau_{21}\tau_{22})$. $d_1^{[1]} = d_1^{[2]} = 0.75$ and $d_2^{[1]} = d_2^{[2]} = 1.25$.

When controllers are not applied, the real and imaginary parts of the state response curves (SRCs) for the error systems

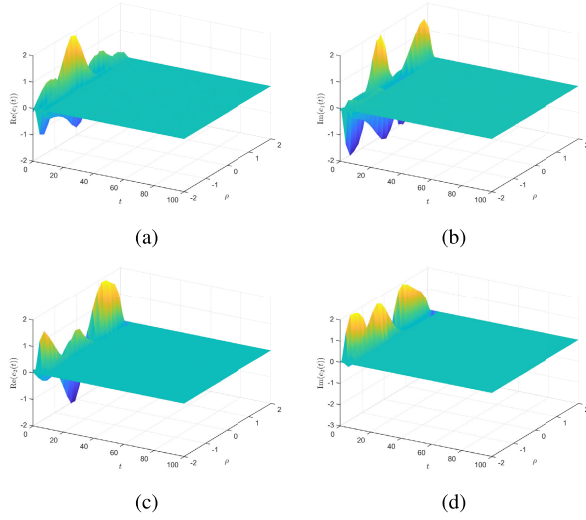


Fig. 4. SRCs of the real and imaginary parts for system (6) under controller (8). (a) $\text{Re}(e_1(t))$, (b) $\text{Im}(e_1(t))$, (c) $\text{Re}(e_2(t))$, and (d) $\text{Im}(e_2(t))$.

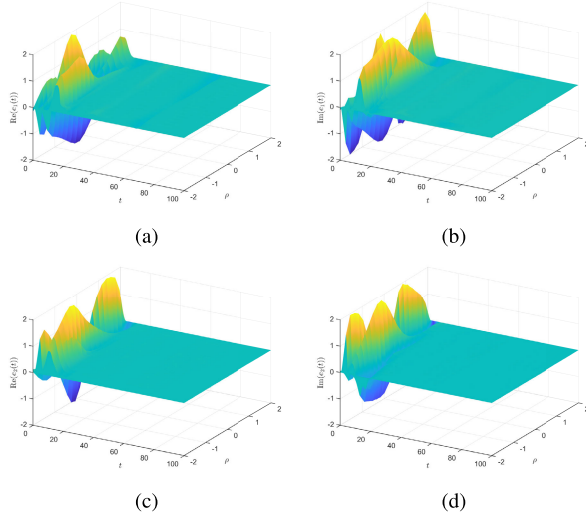


Fig. 5. SRCs of the real and imaginary parts for system (6) under controller (26). (a) $\text{Re}(e_1(t))$, (b) $\text{Im}(e_1(t))$, (c) $\text{Re}(e_2(t))$, and (d) $\text{Im}(e_2(t))$.

(4) and (5) are shown in Fig. 3(a)–(d). It is evident that the error systems (4) and (5) remain desynchronized without the controller.

Case I: We set parameters as $\delta = 1.4$, $\xi_{11} = 0.05$, $\xi_{12} = 0.1$, $\xi_{21} = 0.05$, and $\xi_{22} = 0.1$. According to Theorems 1 and 2, the implementation of adaptive event-triggered controllers (8) drives the error systems (4) and (5) to GPS while excluding Zeno behavior, with the real and imaginary parts of their SRCs of the error systems (4) and (5) characteristics documented in Fig. 4(a)–(d).

Case II: We select identical parameters to Case I, systems (4) and (5) achieve GAS under the adaptive event-triggered controller (27) when $\delta = 0$, as shown in Fig. 5(a)–(d).

Example 2: Consider the 2-D systems (4) and (5) under parameters of Example 1. The calculated values $\|B^\Delta\| = 0.36$, $\|B^*\| = 0.22$, $\|B_1 B_2\| = 5.57$, $\|B_3 B_4\| = 2.76$, $\|C^\Delta\| = 0.21$, $\|C^*\| = 0.16$, $\|C_1 C_2\| = 2.81$, $\|C_3 C_4\| = 1.27$, $\mathcal{W}_\star^* = 40.91$,

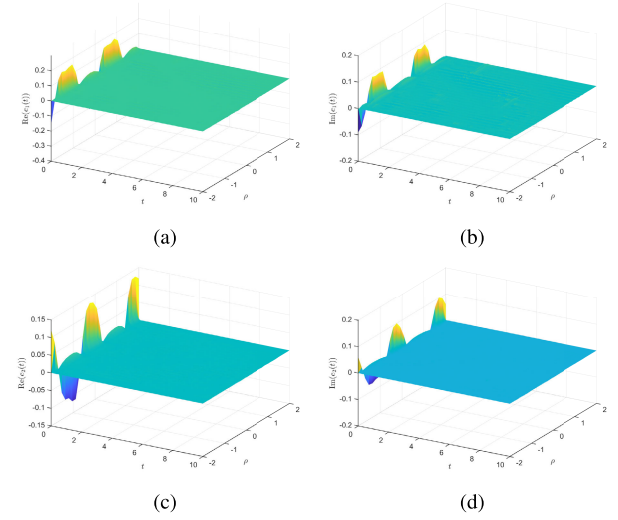


Fig. 6. SRCs of the real and imaginary parts for system (6) under controller (27). (a) $\text{Re}(e_1(t))$, (b) $\text{Im}(e_1(t))$, (c) $\text{Re}(e_2(t))$, and (d) $\text{Im}(e_2(t))$.

$\mathcal{M}_V^* = 2.4$, $\sqcup_1 = -9.95 < 0$, $\sqcup_2 = -0.76 < 0$, which satisfies the conditions of Corollary 2. The real and imaginary characteristics of their SRCs of the error systems (4) and (5) are characterized in Fig. 6(a)–(d).

Remark 7: A comparison of Figs. 4–6 reveals that, given identical model parameters, the system exhibits slightly inferior synchronization performance under the event-triggered controller (28) compared to the adaptive event-triggered controller (8). This indicates the superior effectiveness of the adaptive controller in handling uncertainties.

Example 3: The synchronization results from Example 1 are now applied to chaotic image encryption. An image $\mathbb{B}$ (College.png) of size $M \times N \times 3$ is selected as the plaintext image, and its encryption and decryption process is shown in Fig. 7. Reading the separate color channels of an image (0–255 pixels) produces three pixel intensity matrices: $\mathbb{B}_r(i, j)$ for red, $\mathbb{B}_g(i, j)$ for green, and $\mathbb{B}_b(i, j)$ for blue. The spatial position of each pixel is given by the indices i and j .

- 1) **Image Scrambling:** Two random sequences, $\mathbb{T}_1$ and $\mathbb{T}_2$, each of length $M \times N$, are generated. The sequence $\mathbb{T}_2$ is then used to permute the pixel blocks within the three-channel matrices $\mathbb{B}_r(i, j)$, $\mathbb{B}_g(i, j)$, and $\mathbb{B}_b(i, j)$. Finally, these scrambled channels are combined to produce the disordered image.
- 2) **Image Encryption:** Utilizing the initial values and parameters from Example 1 (Case 1), the drive system first generates the sequence $(\text{Re}(\varphi_2^{[-1]}), \text{Im}(\varphi_2^{[-1]}), \text{Re}(\varphi_2^{[0]}), \text{Im}(\varphi_2^{[0]}))$, where $\varphi_2^{[-1]}$ indicates the complex-valued sequence for state 2 at spatial position -1 and $\varphi_2^{[0]}$ denotes the sequence at spatial position 0. This sequence is then perturbed by $\mathbb{T}_1$ to yield the final driving sequence $(\ell_1, \ell'_1, \ell_2, \ell'_2)$. Based on this, three keys are designed

$$\begin{aligned} r(i) &= (\text{round}(\log_{10}(|\ell_1 - 2\ell'_1 + \ell_2 + 3\ell'_2| + 1) \times 10^4))_{256} \\ g(i) &= (\text{round}(\log_{10}(|2\ell_1 - 3\ell'_1 + 2\ell_2 - \ell'_2| + 1) \times 10^4))_{256} \\ b(i) &= (\text{round}(\log_{10}(|3\ell_1 + \ell'_1 + 2\ell_2 - 2\ell'_2| + 1) \times 10^4))_{256} \end{aligned}$$

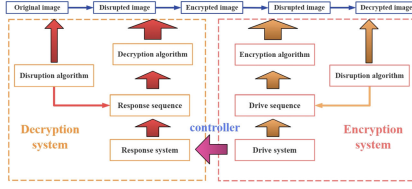
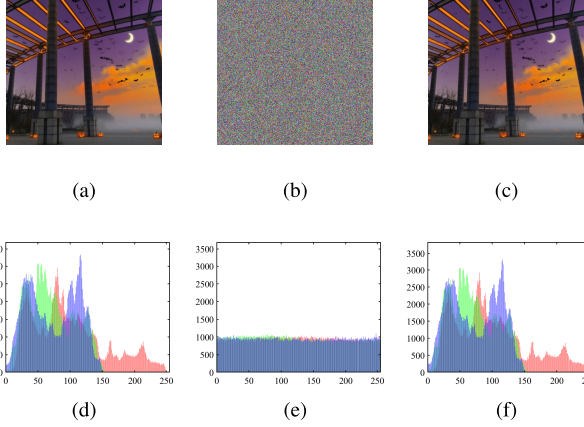


Fig. 7. Encryption and decryption process.

Fig. 8. (a) B_0 , (b) B_1 , (c) B_2 ; histograms, (d) B_0 , (e) B_1 , and (f) B_2 .

where $(\dots)_{256}$ denote $(\dots) \bmod 256$. Subsequently, a segment is captured from ℓ_1 defined by $\ell_1(i) = \ell_1(1 + Loc : M \times N + Loc)$, with $Loc = 10000$. This key sequence, formed by concatenating r , g , and b , is then used to encrypt the original image: each pixel value in the R, G, B channels is XORed with the corresponding key value, resulting in the encrypted image B_1 .

3) *Image Decryption*: The response system sequence is generated and perturbed with T_2 . The encrypted image B_1 is then decrypted by applying the inverse of the encryption process, resulting in the decrypted image B_2 .

The experimental results are shown in Fig. 8(a)–(c). Next, we will further analyze the security performance of the proposed image encryption algorithm.

Image information entropy is used to measure the uncertainty or randomness of pixel values in an image. The formula for calculating information entropy is $\mathcal{H}(\mathbf{w}) = -\sum_{i=0}^{255} \mathfrak{P}(\mathbf{w}_i) \log \mathfrak{P}(\mathbf{w}_i)$, where $\mathfrak{P}(\mathbf{w}_i)$ represents the probability of the grayscale i value appearing in the entire image. The closer the entropy of the ciphertext image is to 8, the stronger the algorithm's ability to conceal original information and resist statistical attacks.

The encryption scheme exhibits excellent performance, as evidenced by the information entropy: the encrypted image reaches 7.9912, which is very close to the ideal value of 8 and a significant increase from the original 7.6161, indicating strong capabilities for concealment and statistical attack resistance. The decrypted image entropy of 7.6212 confirms successful recovery.

Histogram analysis serves to verify if an encryption algorithm can effectively conceal the statistical features of the original image. It is directly related to the algorithm's

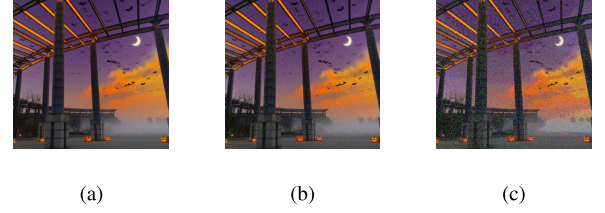
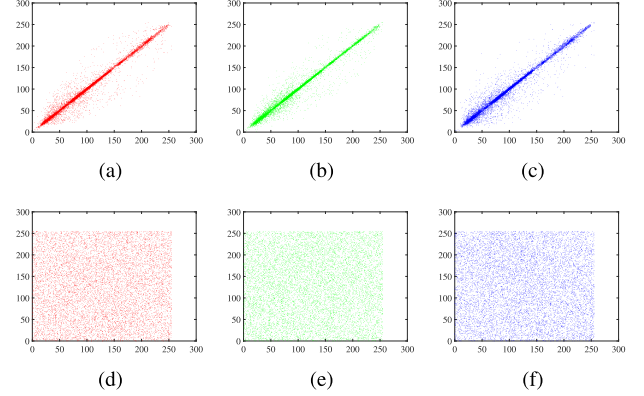
Fig. 9. Synchronous error. (a) 10^{-7} . (b) 10^{-5} . (c) 10^{-4} .

Fig. 10. Correlation of APs in H, V, and D directions. (a)–(c) Original image. (d)–(f) Encrypted image.

TABLE I
ARCHITECTURE OF THE STUDIED NNS MODEL AND TYPE OF CONTROLLER ADOPTED

	[4], [5]	[18]	[20]	[21]	[33]	[41]	[42]	Ours
MNNs	✓	×	×	✓	×	×	×	✓
Delay	B	C	B	B	C	/	C	P
RDNNs	×	✓	✓	×	×	×	×	✓
FNNs	×	✓	✓	✓	✓	×	×	✓
RVNNs	✓	✓	✓	✓	×	×	×	×
CVNNs	×	×	×	×	✓	×	✓	✓

In this table: B stands for bounded time-varying delay, C for constant delay, U for unbounded delay and P for proportional delay.

resistance against statistical attacks, a common cryptanalytic method where attackers use patterns in the ciphertext to deduce information about the plaintext. Thus, a flat and uniform histogram indicates a successful encryption algorithm, showing that it has completely disrupted the original data's statistical patterns and left no useful information in the ciphertext for decryption. Fig. 8(d)–(f) presents a comparison of the histograms of the original image, the encrypted image, and the decrypted image. It can be observed that the histogram of the encrypted image is highly consistent and nearly uniform in distribution. This makes it extremely difficult to break the encrypted image using statistical methods, thereby significantly enhancing security.

Fig. 9(a)–(c) illustrates the image decryption process under different synchronization errors. As the error increases, the decryption quality noticeably degrades. Although such errors cannot be entirely eliminated, they can be minimized to the greatest extent possible.

TABLE II
TYPE OF CONTROLLER ADOPTED

	[4], [5]	[7]	[14]	[19]	[33]	[41]	[42]	Ours
Feedback	✓	×	✓	×	✓	✓	✓	✓
Adaptive	×	✓	×	×	×	✓	✓	✓
Event-triggered	✓	✓	✓	✓	×	×	×	✓

TABLE III
APs CORRELATION IN COLOR IMAGES

Diretion	Image	R	G	B
H	$\mathbb{B}_0$	0.9996	0.9974	0.9978
	$\mathbb{B}_1$	0.0197	0.0008	-0.0135
	$\mathbb{B}_2$	0.9959	0.9762	0.9989
V	$\mathbb{B}_0$	0.9997	0.9983	0.9689
	$\mathbb{B}_1$	-0.0008	0.0126	-0.0053
	$\mathbb{B}_2$	0.9947	0.9869	0.9796
D	$\mathbb{B}_0$	0.9991	0.9951	0.9958
	$\mathbb{B}_1$	0.0152	0.0115	-0.0090
	$\mathbb{B}_2$	0.9950	0.9860	0.9679

In natural images, adjacent pixels (APs) exhibit strong correlations along horizontal (H), vertical (V), and diagonal (D) directions, as quantitatively demonstrated in Table III. These pronounced linear relationships are visually depicted in Fig. 10(a)–(c). The performance of an encryption algorithm can be evaluated by calculating and comparing the correlation coefficients of such pixel pairs before and after encryption. Our method successfully disrupts these spatial correlations, reducing the coefficients to values near zero, as shown in Fig. 10(d)–(f). This indicates effective obscuration of statistical patterns and enhanced resistance against statistical attacks, thereby ensuring strong confidentiality.

V. CONCLUSION

This work explores GPS for uncertain complex-valued reaction-diffusion T-S fuzzy MNNs with proportional delays via adaptive event-triggered control. Two numerical simulations validate the theoretical conclusions and demonstrate that the adaptive event-triggered controller achieves superior synchronization performance compared to the conventional event-triggered controller. In particular, the rationality of applying Example 1 to image decryption is verified by analyzing the information entropy, histogram, and APs correlations of the encrypted image. However, limitations of this work include the requirement of bounded activation functions and consideration only for RDNNs with 1-D spatial domains. Future work will focus on addressing the unboundedness of activation functions under weaker assumptions, extending the proposed control scheme to higher-dimensional spatial domains for better practical alignment, and applying the proposed methodology to NNs with other types of unbounded delays.

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